1 (a) Systematic errors will result in all readings being either above or below the accepted value. B1

Random errors will result in readings being scattered around the accepted value (ie equal probability being above or below the accepted value)

Systematic errors cannot be eliminated by repeating readings and then averaging. B1

Random errors can be reduced by repeating the readings and averaging.

Systematic error: same magnitude and sign

Random error: different magnitude and sign

(b)(i) $T = 2\pi\sqrt{(\frac{0.50}{9.8})} = 1.419$ A1

$\frac{\Delta T}{T} = \frac{1}{2}(\frac{\Delta L}{L} + \frac{\Delta g}{g})$ M1

$\Rightarrow \Delta T = \frac{1}{2}(\frac{0.2}{50.0} + \frac{0.1}{9.8}) \times 1.419 = 0.01$ (to 1 s.f)

$T = 1.42 \pm 0.01$ s A1

(b)(ii) Due to human reaction time, large number of oscillations will reduce the fractional or percentage uncertainty of the calculated value of T. B1

2 (a) The gradient of the graph gives acceleration. B1